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(54) **HOUSING FOR ENERGY METER**

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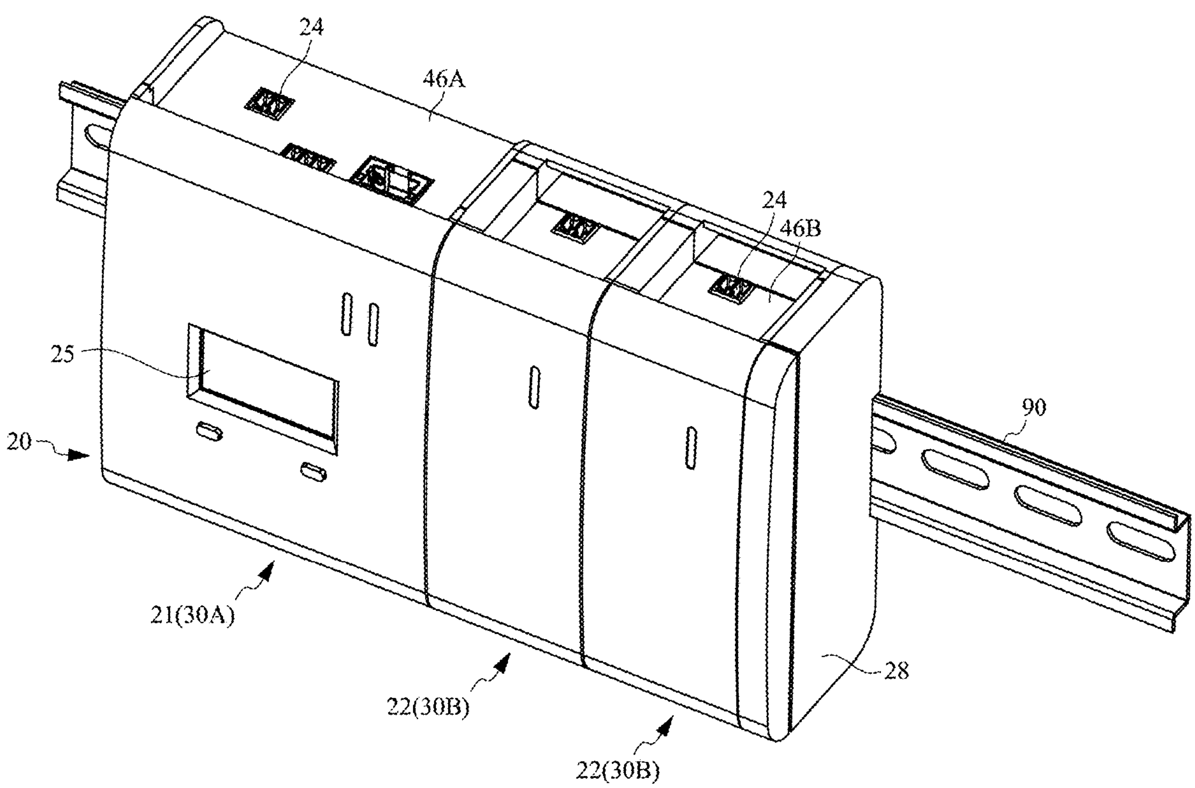
(57) **ABSTRACT**

A housing for an energy meter includes a body, a cover, a chute structure, and a slider. The body has a first surface and a vertical wall substantially perpendicular to the first surface. The cover has a side wall. The chute structure includes a chute and a peripheral wall surrounding the chute. The peripheral wall is provided with a limiting element. The chute structure is disposed on one of the vertical wall and the side wall, and the slider is disposed on the other one of the vertical wall and the side wall. The slider is slidably disposed in the chute. The limiting element is configured to limit a position of the slider. The cover is slidably disposed on the body through the slider and the chute structure.



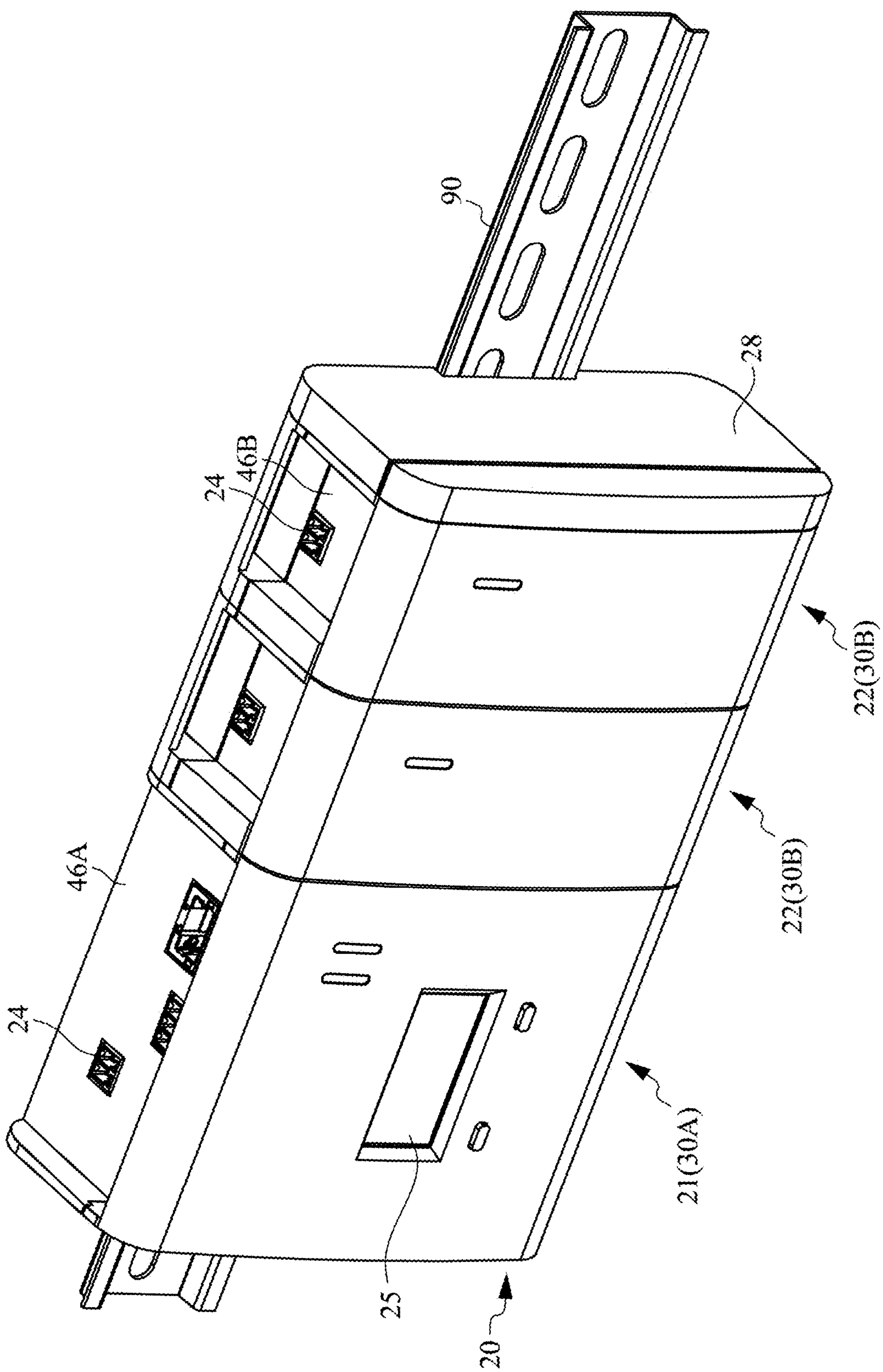


Fig. 1

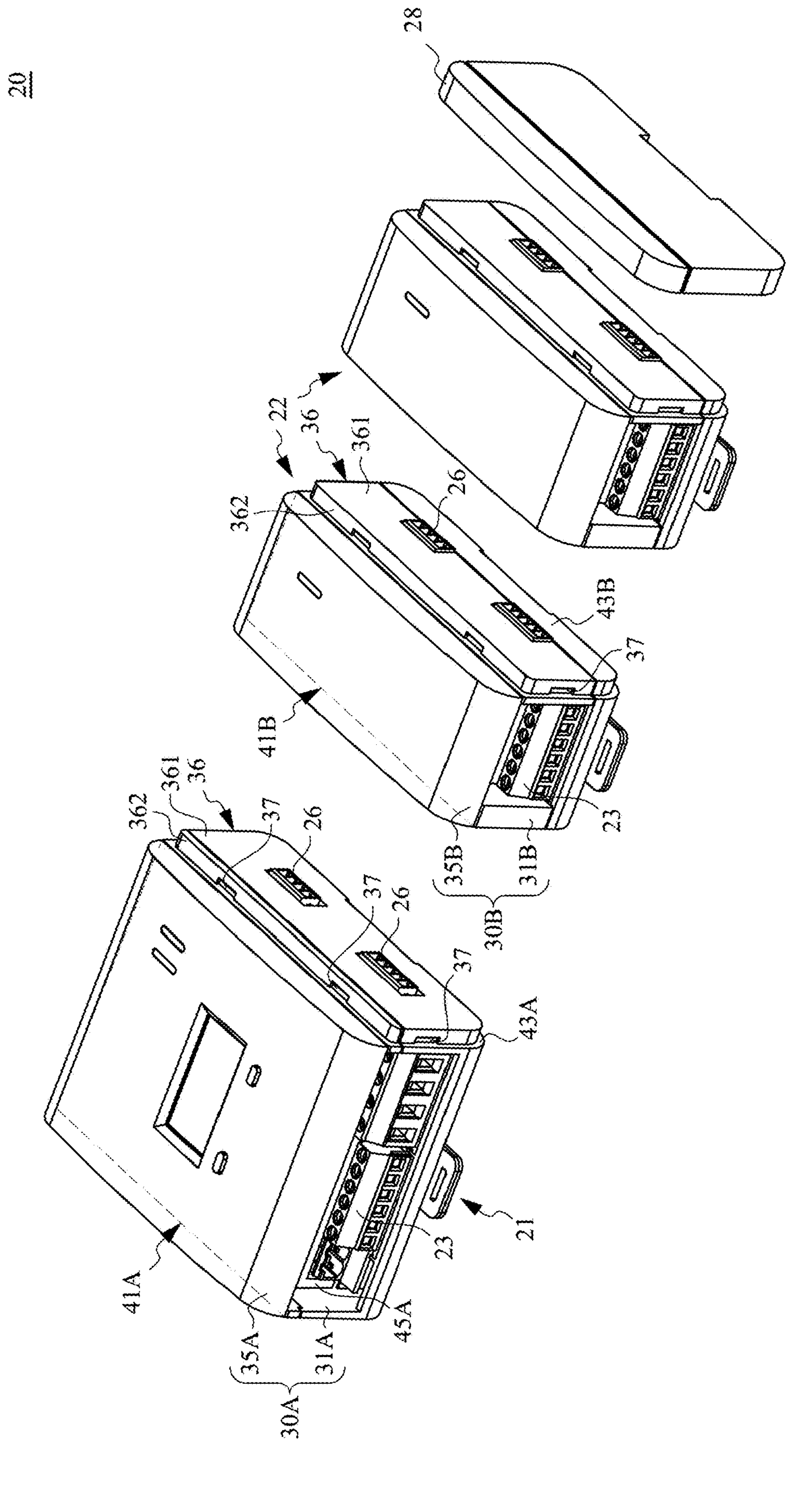

20
28
22
36
361
362
26
43B
37
41B
23
35B
31B
30B
362
361
36
37
26
37
26
37
43A
41A
23
21
35A
31A
45A
30A

Fig. 2

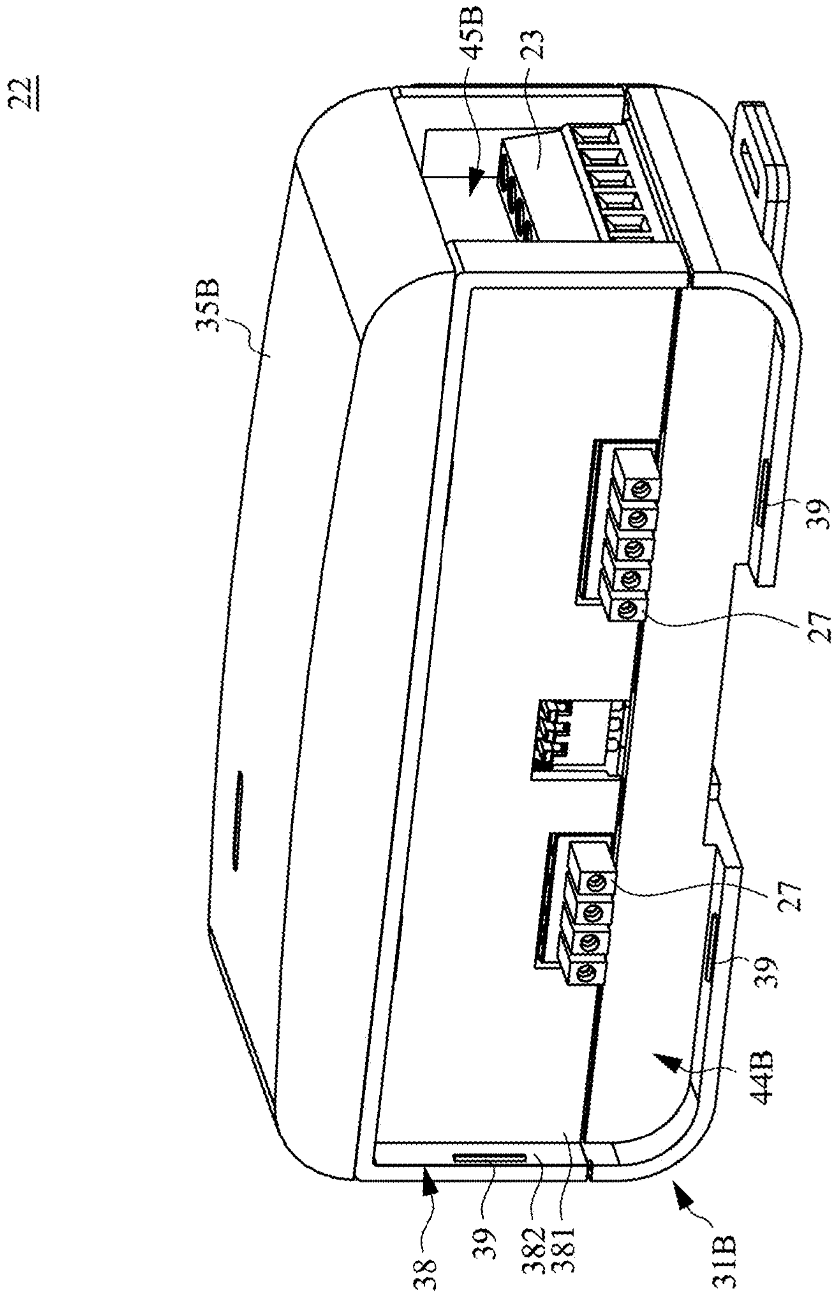


Fig. 3

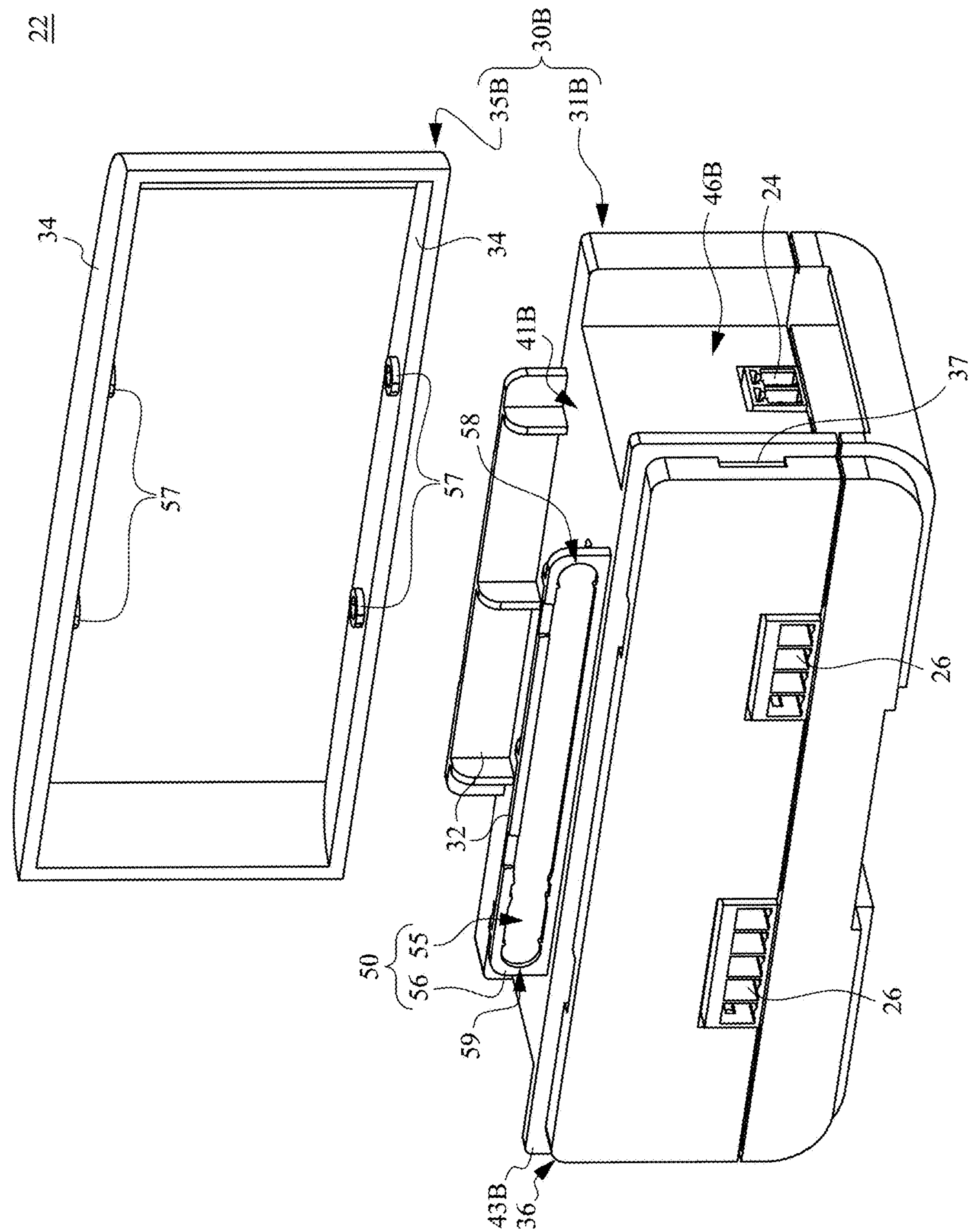

22
30B
35B
31B
34
34
46B
24
41B
37
58
57
57
26
26
32
50
55
56
59
43B
36

Fig. 4

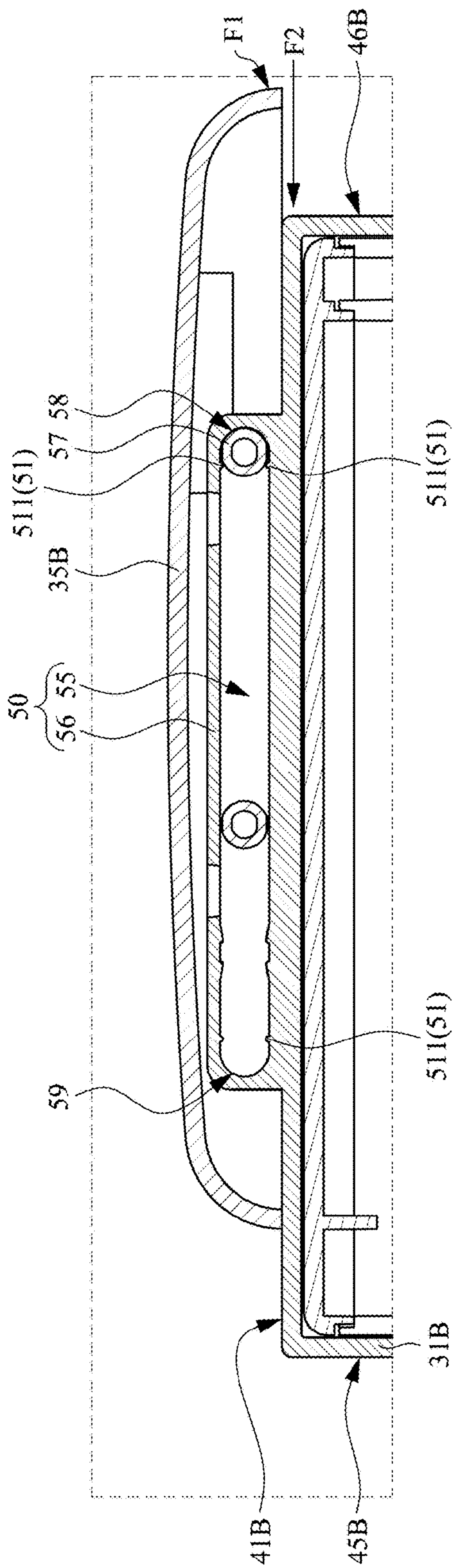


Fig. 5

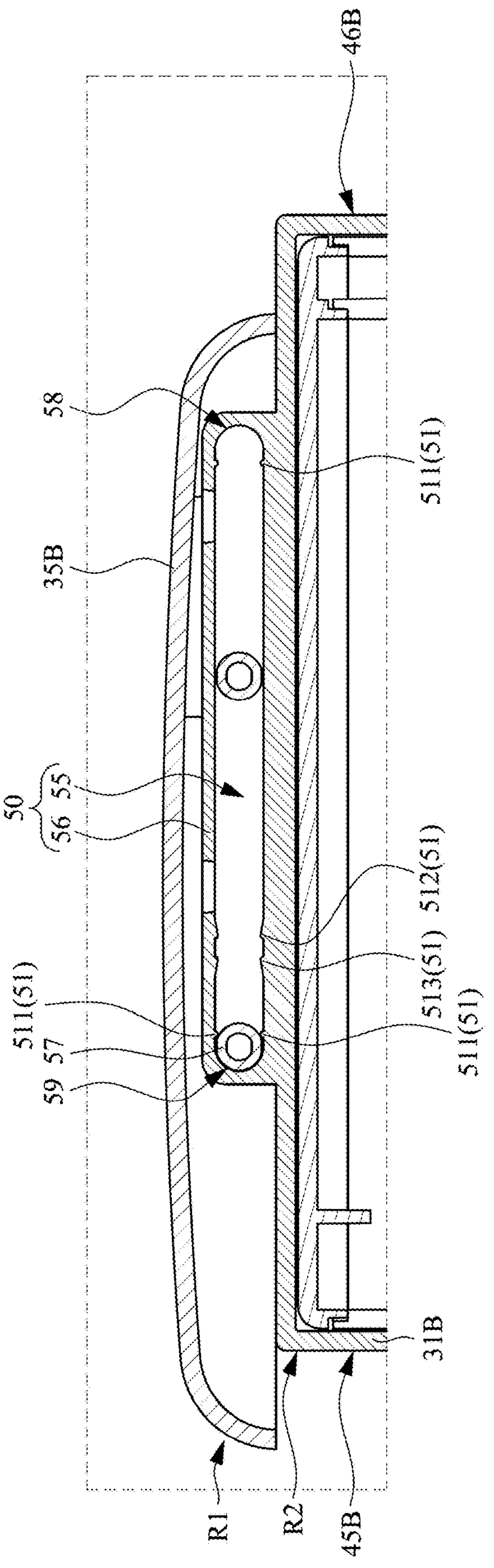

46B
58
511(51)
35B
50
55
56
512(51)
513(51)
511(51)
57
59
511(51)
31B
R1
R2
45B

Fig. 6

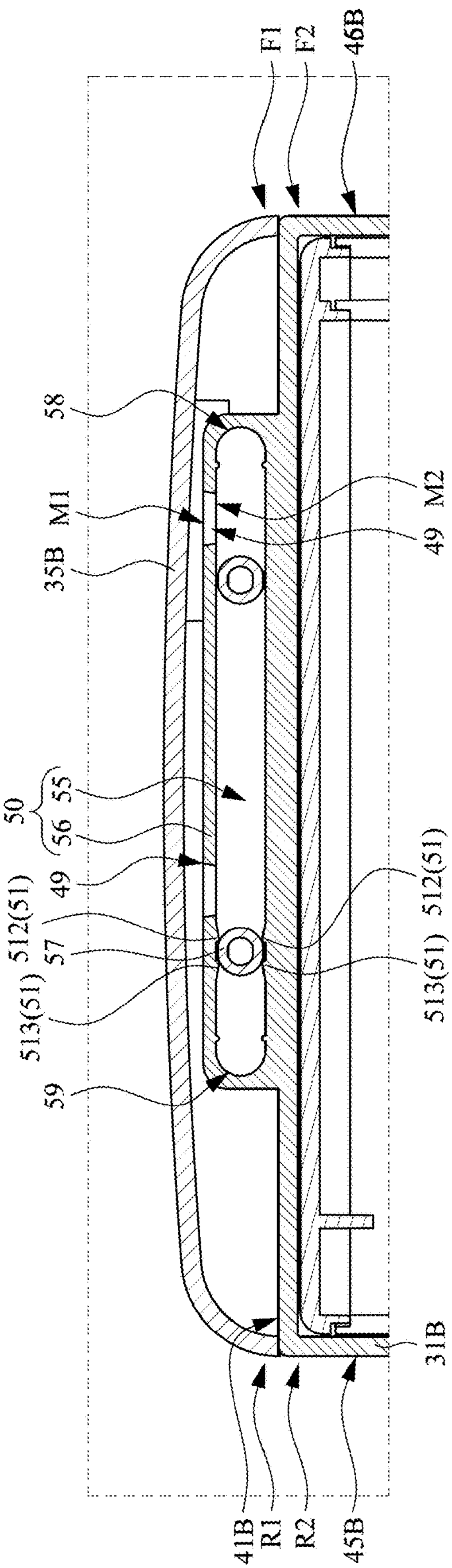

F1
F2
46B
58
M1
35B
M2
49
50
55
56
49
512(51)
57
513(51)
512(51)
513(51)
59
41B
R1
R2
45B
31B

Fig. 7

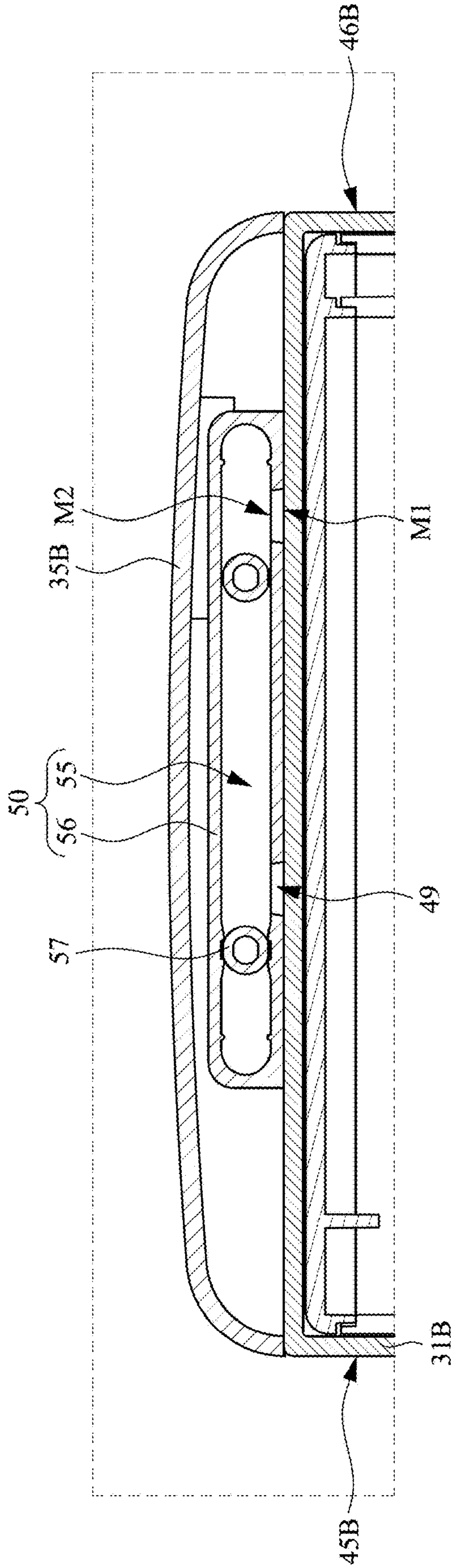


Fig. 8

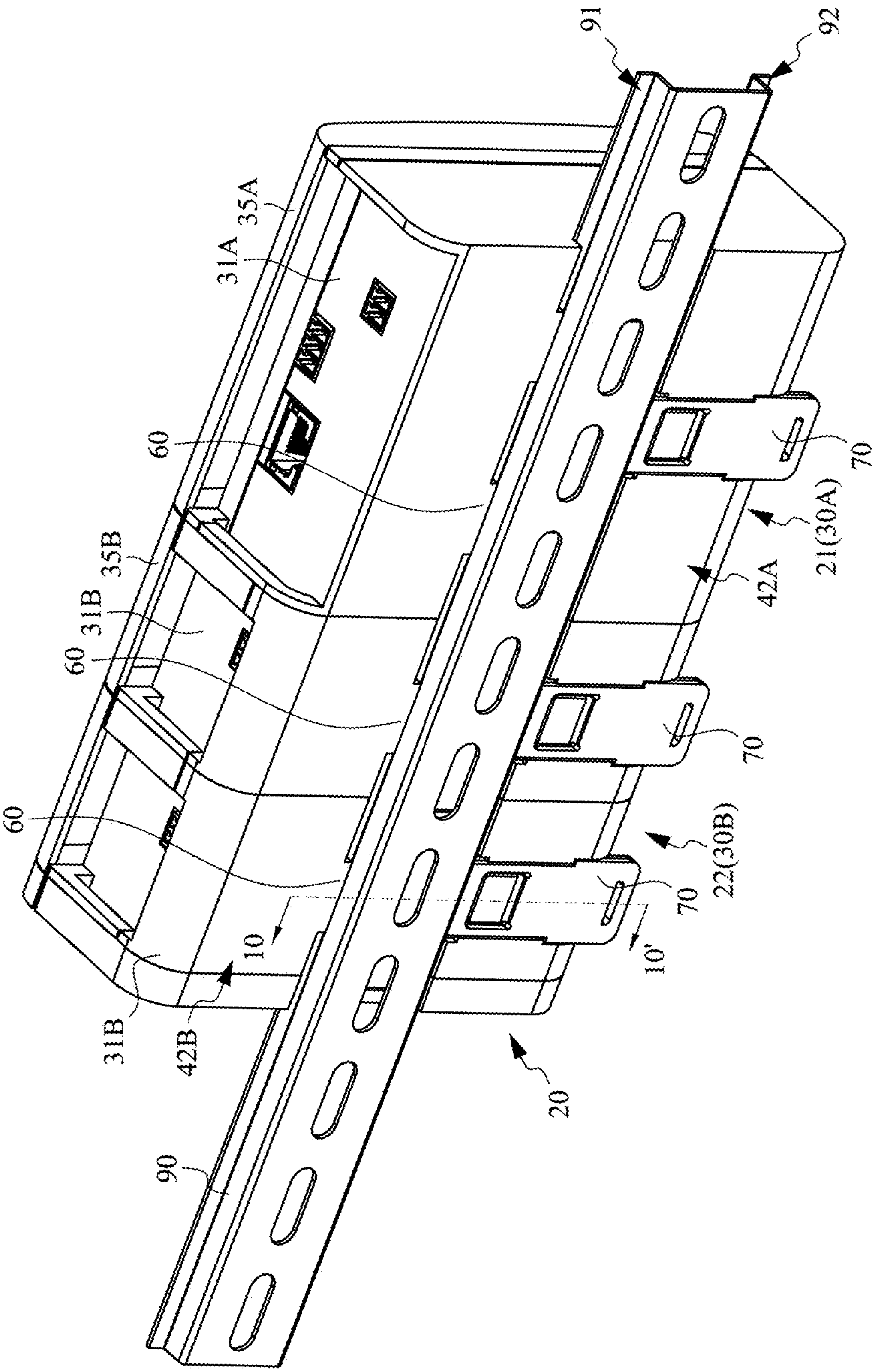

91
92
31A
35A
60
70
21(30A)
42A
35B
31B
60
70
22(30B)
60
70
10
10'
31B
42B
20
90

Fig. 9

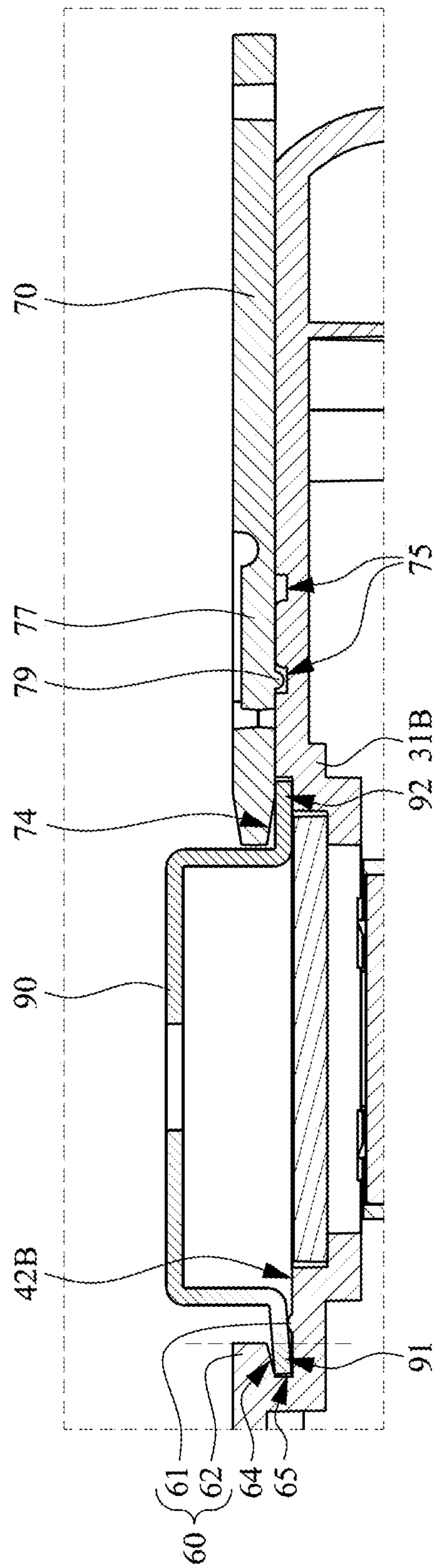


Fig. 10

22

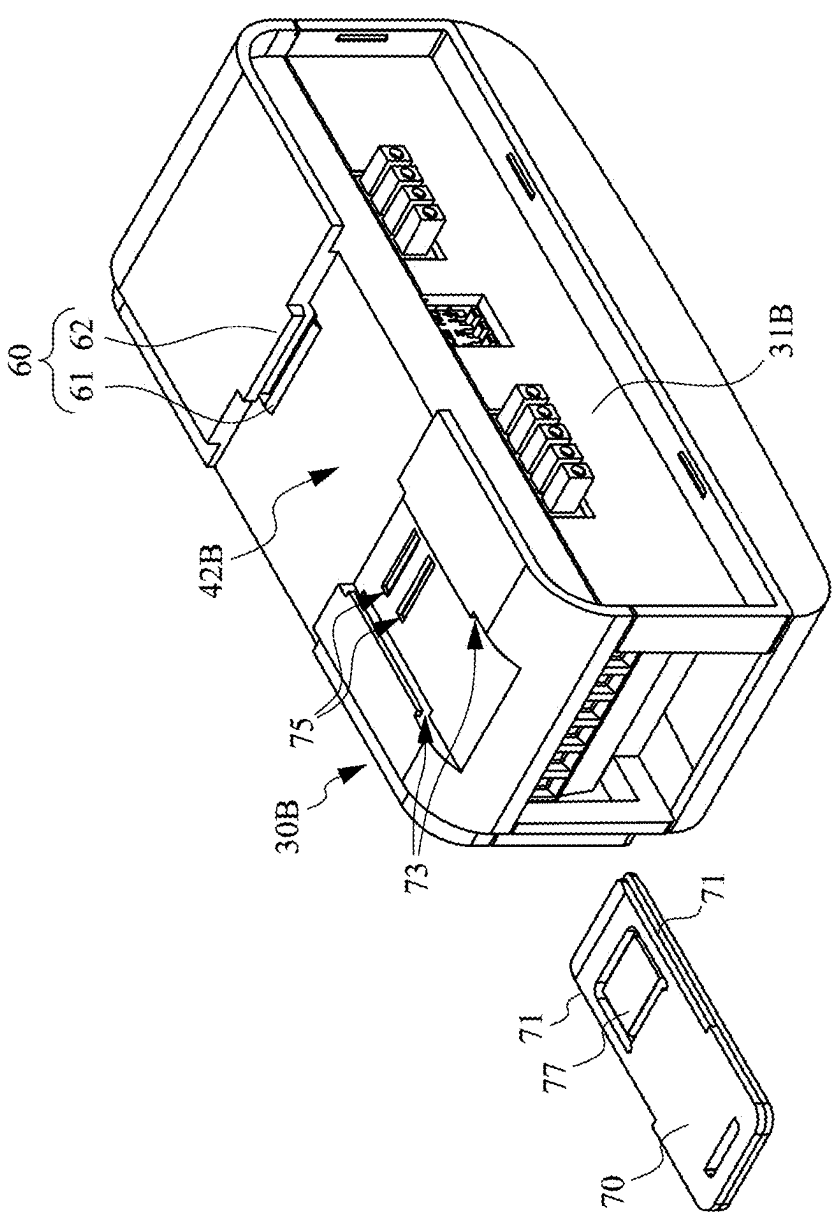


Fig. 11

70

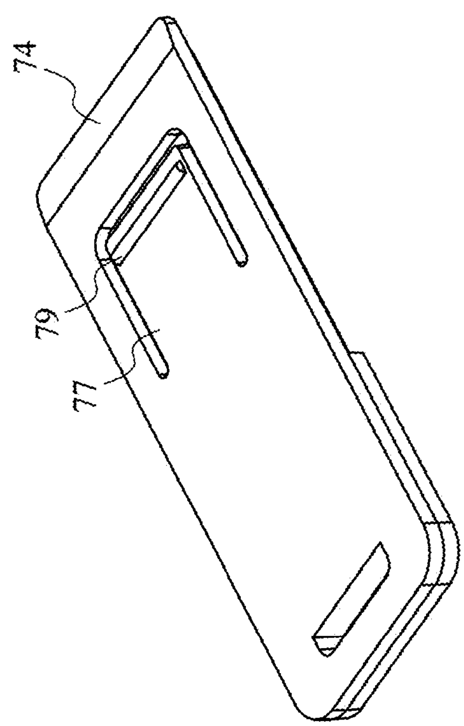


Fig. 12

HOUSING FOR ENERGY METER

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority to Taiwan Patent Application No. 113107624, filed on Mar. 1, 2024, the disclosure of which is incorporated herein by reference in its entirety.

FIELD OF INVENTION

[0002] The present disclosure relates to a housing for an energy meter.

BACKGROUND

[0003] In recent years, a trend of energy conservation and carbon reduction has gradually increased application of smart power meters. However, wiring and installation of current smart power meters are inconvenient, and the current smart power meters are not expandable.

SUMMARY

[0004] In order to solve the above technical problem, the present disclosure provides a housing for an energy meter, which includes a body, a cover, a chute structure, and a slider. The body has a first surface and a vertical wall substantially perpendicular to the first surface. The cover has a side wall. The chute structure includes a chute and a peripheral wall surrounding the chute. The peripheral wall is provided with a first limiting element. The chute structure is disposed on one of the vertical wall and the side wall, and the slider is disposed on the other one of the vertical wall and the side wall. The slider is slidably disposed in the chute. The first limiting element is configured to limit a position of the slider. The cover is slidably disposed on the body through the slider and the chute structure.

[0005] In an embodiment of the present disclosure, the first limiting element is close to an end of the chute. When the slider is limited between the first limiting element and the end of the chute, a front edge of the cover is in front of a front edge of the body, or a rear edge of the cover is behind a rear edge of the body.

[0006] In an embodiment of the present disclosure, the peripheral wall of the chute structure is further provided with a second limiting element and a third limiting element. When the slider is limited between the second limiting element and the third limiting element, a front edge of the cover is substantially aligned with a front edge of the body, or a rear edge of the cover is substantially aligned with a rear edge of the body.

[0007] In an embodiment of the present disclosure, the peripheral wall is further provided with a notch, and the notch communicates with the chute and is configured to allow the slider to pass.

[0008] In an embodiment of the present disclosure, the notch has a first opening at an outer surface of the peripheral wall and a second opening at an inner surface of the peripheral wall, and a width of the first opening is greater than a width of the second opening.

[0009] In an embodiment of the present disclosure, the body further comprises a second surface, and the second surface is provided with a holding structure. The holding structure comprises a first protrusion and a second protrusion. The first protrusion is disposed on the second surface. The second protrusion extends from the second surface and forms a space with the second surface. The space is configured to receive a first edge of a bracket. The first protrusion is configured to abut against the first edge such that the second protrusion interferes with the first edge.

[0010] In an embodiment of the present disclosure, the second protrusion has a first inclined surface, and the first inclined surface faces the second surface and is configured to interfere with the first edge of the bracket.

[0011] In an embodiment of the present disclosure, a projection of the second protrusion on the second surface does not overlap with a projection of the first protrusion on the second surface.

[0012] In an embodiment of the present disclosure, the housing further includes a holding element, a guide rail, and a guide groove. The guide rail is disposed on one of the holding element and the second surface, and the guide groove is disposed on the other one of the holding element and the second surface. The guide rail is slidably disposed in the guide groove, so that the holding element is slidably disposed on the second surface. The holding element is configured to hold a second edge of the bracket.

[0013] In an embodiment of the present disclosure, a front end of the holding element has a second inclined surface, and the second inclined surface faces the second surface and is configured to interfere with the second edge of the bracket.

[0014] In an embodiment of the present disclosure, the second surface is further provided with a first engaging element. The holding element has an elastic arm. The elastic arm is provided with a second engaging element. The second engaging element is configured to engage with the first engaging element to fix a position of the holding element on the second surface.

[0015] In an embodiment of the present disclosure, the body further comprises a third surface. The third surface is provided with a boss, and an outer wall of the boss is provided with a first latching element.

[0016] In an embodiment of the present disclosure, the body further comprises a fourth surface. The fourth surface is provided with a recess that matches the boss, and an inner wall of the recess is provided with a second latching element that matches the first latching element.

[0017] In sum, a housing of an energy meter of the present disclosure includes a body and a cover slidably disposed on the body. The cover can slide upward or downward relative to the body to expose a terminal block and/or a connector disposed on the body, thereby facilitating a user to connect external devices to the energy meter through wires. Furthermore, the cover is slidably disposed on the body through one or more sliders and one or more chute structures. Each of the chute structures includes a chute and a peripheral wall surrounding the chute. The peripheral wall is provided with one or more limiting elements configured to limit positions of the sliders in the chute, so that the sliders can stay at specific positions. Moreover, a back surface of the body may be provided with a holding structure and a slidable holding element to facilitate the user to mount the energy meter on a bracket. In addition, a side surface of the body may be provided with a boss or a recess. Through a combination of the boss and the recess, multiple energy meters can be combined into an energy meter assembly.

BRIEF DESCRIPTION OF DRA WINGS

[0018] FIG. **1** is a schematic diagram showing an energy meter assembly installed on a bracket according to a first embodiment of the present disclosure.

[0019] FIG. **2** is an exploded view of the energy meter assembly in FIG. **1**.

[0020] FIG. **3** is a schematic diagram of a second energy meter in FIG. **2** viewed from another angle.

[0021] FIG. **4** is an exploded view of the second energy meter in FIG. **3**.

[0022] FIG. **5** to FIG. **7** are partial cross-sectional views of the second energy meter in FIG. **3**, with a cover in different positions.

[0023] FIG. **8** is a partial cross-sectional view of the second energy meter according to a second embodiment of the present disclosure.

[0024] FIG. **9** is a schematic diagram of the energy meter assembly and the bracket in FIG. **1** viewed from another angle.

[0025] FIG. **10** is a cross-sectional view of the energy meter assembly and the bracket in FIG. **9** taken at a line **10-10'**.

[0026] FIG. **11** is a schematic diagram showing the second energy meter in FIG. **9** after removing a holding element.

[0027] FIG. **12** is a schematic diagram of the holding element in FIG. **11** viewed from another angle.

DETAILED DESCRIPTION

[0028] In order to make the description of the present disclosure more detailed and complete, reference may be made to the accompanying drawings and various embodiments described below. Elements in the drawings are not drawn to scale and are provided solely to illustrate the present disclosure. Many practical details are described below to provide a comprehensive understanding of the present disclosure. However, one of ordinary skill in the relevant art will understand that the present disclosure may be practiced without one or more of the practical details. Accordingly, these details should not be used to limit the present disclosure.

[0029] Please refer to FIG. **1**, which is a schematic diagram showing an energy meter assembly **20** installed on a bracket **90** according to a first embodiment of the present disclosure. The energy meter assembly **20** includes a first energy meter **21** and one or more second energy meters **22**. One of the second energy meters **22** is detachably connected to the first energy meter **21**, and any two adjacent second energy meters **22** are detachably connected to each other. The first energy meter **21** and the one or more second energy meters **22** are configured to acquire energy usage information from various devices. The energy usage information includes one or more of gasoline usage information, electricity usage information, water usage information, and gas usage information.

[0030] Please refer to FIG. **1**, the first energy meter **21** is a master energy meter, and the one or more second energy meters **22** are slave energy meters. Taking the energy meter assembly **20** including one first energy meter **21** and one second energy meter **22** as an example, the first energy meter **21** may be configured to: acquire first energy usage information from a device, acquire second energy usage information from the second energy meter **22**, count and analyze the first and second energy usage information, calculate carbon emission based on the first and second energy usage information, transmit the first and second energy usage information to an electronic device (for example, transmit the first and second energy usage information to a users' mobile device through Bluetooth communication, or transmit the first and second energy usage information to a server through network), and display the first and second energy usage information on a display **25** of the first energy meter **21**. The second energy meter **22** may be configured to: acquire the second energy usage information from another device, and transmit the second energy usage information to the first energy meter **21**.

[0031] Please refer to FIG. **2**, which is an exploded view of the energy meter assembly **20** in FIG. **1**. The first energy meter **21** includes a first housing **30**A. The first housing **30**A includes a first body **31**A and a first cover **35**A. The first cover **35**A is disposed on the first body **31**A, and covers a first surface **41**A (shown in dotted lines) of the first body **31**A. The second energy meter **22** includes a second housing **30**B. The second housing **30**B includes a second body **31**B and a second cover **35**B. The second cover **35**B is disposed on the second body **31**B, and covers a first surface **41**B (shown in dotted lines) of the second body **31**B. The first housing **30**A and the second housing **30**B may be made of insulating materials, such as plastic, but are not limited thereto.

[0032] Please refer to FIG. **2**, a third surface **43**A of the first body **31**A of the first energy meter **21** is provided with a boss **36**. The boss **36** has a top surface **361** and an outer wall **362** surrounding the top surface **361**. An outer surface of the outer wall **362** is provided with one or more first latching elements **37**. Please refer to FIG. **3**, which is a schematic diagram of the second energy meter **22** in FIG. **2** viewed from another angle. A fourth surface **44**B of the second body **31**B of the second energy meter **22** is provided with a recess **38**. The recess **38** has a bottom surface **381** and an inner wall **382** surrounding the bottom surface **381**. An inner surface of the inner wall **382** is provided with one or more second latching elements **39**. The recess **38** corresponds to and matches the boss **36**. That is, the recess **38** has a complementary shape to the boss **36**. The second latching elements **39** correspond to and match the first latching elements **37**. That is, the second latching elements **39** have a complementary shape to the first latching elements **37**.

[0033] Please refer to FIG. **2**, the first energy meter **21** may further include one or more first connectors **26** embedded in the top surface **361** of the boss **36**. Please refer to FIG. **3**, the second energy meter **22** may further include one or more second connectors **27** embedded in the bottom surface **381** of the recess **38**. By combining the boss **36** with the recess **38**, and engaging the second latching elements **39** with the first latching elements **37**, the second energy meter **22** is detachably connected to the first energy meter **21**. At this time, the second connectors **27** are connected to the first connectors **26**. When the first and second connectors **26** and **27** are connected, the first energy meter **21** can communicate with the second energy meter **22** and supply power to the second energy meter **22**, and the second energy meter **22** can transmit the second energy usage information to the first energy meter **21**.

[0034] As shown in FIG. **2** and FIG. **3**, in this embodiment, the first latching elements **37** are engaging slots, and the second latching elements **39** are engaging blocks. In other embodiments, the first latching elements **37** may be engaging blocks, and the second latching elements **39** may be engaging slots. In this embodiment, a plurality of first latching elements **37** are spaced around the boss **36**, and a plurality of second latching elements **39** are spaced around the recess **38**. In other embodiments, one or more sides of the boss **36** may be provided with one or more first latching elements **37**, and one or more sides of the recess **38** may be provided with one or more second latching elements **39**.

[0035] As shown in FIG. **2**, a third surface **43**B of the second body **31**B of the second energy meter **22** is also provided with the boss **36**, the first latching elements **37**, and the first connectors **26**, so as to facilitate series connection of two or more second energy meters **22**. The third surface **43**B and the fourth surface **44**B are two opposite surfaces of the second body **31**B. Specifically, by combining the boss **36** of one second energy meter **22** with the recess **38** of another second energy meter **22**, and engaging the second latching elements **39** in the recess **38** with the first latching elements **37** on the boss **36**, the aforementioned two second energy meters **22** are detachably connected.

[0036] In other embodiments, the third surface **43**A of the first body **31**A of the first energy meter **21** may be provided with the recess **38** and the one or more second latching elements **39** in the recess **38**, the fourth surface **44**B of the second body **31**B of each second energy meter **22** may be provided with the boss **36** and the one or more first latching elements **37** on the boss **36**, and the third surface **43**B of the second body **31**B may be provided with the recess **38** and the one or more second latching elements **39** in the recess **38**.

[0037] Please refer to FIG. **1** and FIG. **2**, in an embodiment, the energy meter assembly **20** further includes an end cover **28**. The end cover **28** is provided with a recess (not shown) that corresponds to and matches the boss **36**. By combining the recess of the end cover **28** with the boss **36** of the first energy meter **21** or the boss **36** of the second energy meter **22**, the end cover **28** covers the third surface **43**A of the first energy meter **21** or the third surface **43**B of the second energy meter **22** that is farthest from the first energy meter **21**, thereby covering the unused first connectors **26**. The recess of the end cover **28** may also be provided with the one or more second latching elements (not shown) configured to engage the first latching elements **37** on the boss **36**.

[0038] Please refer to FIG. **4**, which is an exploded view of the second energy meter **22** in FIG. **3**. The second body **31**B of the second energy meter **22** further includes one or more vertical walls **32**. Each of the vertical walls **32** is provided with a chute structure **50**. In this embodiment, the second body **31**B includes two vertical walls **32**, and the two vertical walls **32** are located on two opposite sides of the first surface **41**B and are substantially perpendicular to the first surface **41**B. Specifically, the two vertical walls **32** are respectively close to the third surface **43**B and the fourth surface **44**B of the second body **31**B. Each of the chute structures **50** is disposed on a surface of the corresponding vertical wall **32** away from the other vertical wall **32**. Each of the chute structures **50** includes a chute **55** and a peripheral wall **56** surrounding the chute **55**.

[0039] As shown in FIG. **4**, the second cover **35**B of the second energy meter **22** has one or more side walls **34**. An inner surface of each of the side walls **34** is provided with one or more sliders **57**. In this embodiment, the sliders **57** are disposed on the two side walls **34** on two opposite sides of the second cover **35**B, that is, they are disposed on the two side walls **34** corresponding to the two chute structures **50**. Each of the side walls **34** is provided with two sliders **57**, but is not limited thereto. The sliders **57** correspond to the chutes **55** of the chute structures **50** and are slidably disposed in the chutes **55**. The second cover **35**B is slidably disposed on the second body **31**B through the sliders **57** and the chute structures **50**. In this embodiment, the sliders **57** are shaped as circles, but are not limited thereto. The sliders **57** may be shaped as ellipses, polygons, or irregular shapes, as long as the sliders **57** can be slidably disposed in the chutes **55**.

[0040] As shown in FIG. **4**, each of the chutes **55** has a first end **58** and a second end **59** opposite to each other, and the sliders **57** are configured to slide between the first end **58** and the second end **59** of the chute **55**. Please refer to FIG. **5**, an inner surface of the peripheral wall **56** of each of the chute structures **50** may be provided with one or more limiting elements **51**. The limiting elements **51** are configured to limit positions of the sliders **57**. Specifically, the limiting elements **51** are configured to block the sliders **57** from sliding in the chute **55**, so that the sliders **57** stay at specific positions in the chute **55**. A certain external force needs to be applied to deform the sliders **57** or the limiting elements **51** in order for the sliders **57** to pass through the limiting elements **51** and continue to slide.

[0041] Please refer to FIG. **5**, the limiting elements **51** may include one or more first limiting elements **511**, which are close to the first end **58** and/or the second end **59** of the chute **55**. In this embodiment, the first limiting elements **511** are hemispherical protrusions, but are not limited thereto. The first limiting elements **511** may be protrusions with other shapes, as long as the first limiting elements **511** can limit the positions of the sliders **57**.

[0042] Please refer to FIG. **5**, in this embodiment, the peripheral wall **56** is provided with two first limiting elements **511** aligned up and down near the first end **58** of the chute **55**. In other embodiments, the peripheral wall **56** may be provided with only one first limiting element **511**, which is located at an upper edge or a lower edge of the peripheral wall **56** near the first end **58**. When the slider **57** is limited between the first limiting element **511** and the first end **58** of the chute **55**, a front edge F1 of the second cover **35**B is in front of a front edge F2 of the second body **31**B, and the second cover **35**B cannot slide.

[0043] Please refer to FIG. **3**, the second body **31**B of the second energy meter **22** has a fifth surface **45**B. The second energy meter **22** further includes a terminal block **23**. The terminal block **23** is located between the second body **31**B and the second cover **35**B and close to the fifth surface **45**B. The terminal block **23** is configured to connect one or more current transformers and/or flow meters. When the second cover **35**B slides to the position shown in FIG. **5**, the second cover **35**B does not cover the terminal block **23**. That is, the terminal block **23** is exposed, so as to facilitate the user to connect the current transformers and/or the flow meters to the terminal block **23** through wires.

[0044] Please refer to FIG. **6**, in this embodiment, the peripheral wall **56** is provided with two first limiting elements **511** aligned up and down near the second end **59** of the chute **55**. In other embodiments, the peripheral wall **56** may be provided with only one first limiting element **511**, which is located at the upper edge or the lower edge of the peripheral wall **56** near the second end **59**. When the slider **57** is limited between the first limiting element **511** and the second end **59** of the chute **55**, a rear edge R1 of the second cover **35**B is behind a rear edge R**2** of the second body **31**B, and the second cover **35**B cannot slide.

[0045] Please refer to FIG. **4**, the second body **31**B of the second energy meter **22** has a sixth surface **46**B. The sixth surface **46**B and the fifth surface **45**B are two opposite surfaces of the second body **31**B. The second energy meter **22** further includes a third connector **24**. The third connector **24** is disposed between the second body **31**B and the second cover **35**B, is embedded in the sixth surface **46**B, and is configured to connect network equipment or other communication equipment. When the second cover **35**B slides to the position shown in FIG. **6**, the second cover **35**B does not cover the third connector **24**. That is, the third connector **24** is exposed, so as to facilitate the user to connect the network equipment or other communication equipment to the third connector **24** through wires.

[0046] Please refer to FIG. **7**, the limiting elements **51** may further include one or more second limiting elements **512** and one or more third limiting elements **513**. In this embodiment, the peripheral wall **56** is provided with two second limiting elements **512** aligned up and down and two third limiting elements **513** aligned up and down. In other embodiments, the peripheral wall **56** may be provided with one second limiting element **512** located at the upper edge or the lower edge of the peripheral wall **56**, and one third limiting element **513** located at the upper edge or the lower edge of the peripheral wall **56**. When the slider **57** is limited between the second limiting element **512** and the third limiting element **513**, the front edge F**1** of the second cover **35**B is substantially aligned with the front edge F**2** of the second body **31**B, and/or the rear edge R**1** of the second cover **35**B is substantially aligned with the rear edge R**2** of the second body **31**B, so that the second cover **35**B covers the terminal block **23** and/or the third connector **24**, and the second cover **35**B cannot slide.

[0047] Please refer to FIG. **7**, the peripheral walls **56** may be provided with one or more notches **49** communicating with the chute **55**. The notches **49** are configured to allow the sliders **57** to pass. Specifically, the sliders **57** are installed into the chute **55** through the notches **49**. The number of notches **49** is the same as the number of sliders **57**, and positions of the notches **49** correspond to the positions of the sliders **57**.

[0048] Please refer to FIG. **7**, the notches **49** are provided on an edge of the peripheral wall **56** away from the second body **31**B. Each of the notches **49** has a first opening M**1** at an outer surface of the peripheral wall **56** and a second opening M**2** at an inner surface of the peripheral wall **56**. A width of the first opening M**1** is greater than a width of the second opening M**2**. That is, each of the notches **49** tapers in a direction from the peripheral wall **56** to the chute **55**. In other words, an inner surface of the notch **49** is an inclined surface for guiding the slider **57** to enter the chute **55** through the notch **49**.

[0049] Please refer to FIG. **7**, in an embodiment, the width of the second opening M**2** of the notch **49** is slightly less than a maximum width of the slider **57**. After the slider **57** enters the chute **55**, a certain external force needs to be applied to deform the slider **57** slightly so that the slider **57** can pass through the second opening M**2** and leave the chute **55** through the first opening M**1**.

[0050] Please refer to FIG. **8**, which is a partial cross-sectional view of the second energy meter **22** according to a second embodiment of the present disclosure. The second embodiment differs from the first embodiment in that: the one or more sliders **57** are provided on the one or more vertical walls **32** of the second body **31**B, the one or more chute structures **50** are provided on the one or more side walls **34** of the second cover **35**B, and the one or more notches **49** are provided on an edge of the peripheral wall **56** close to the second body **31**B. In this embodiment, two chute structures **50** are respectively provided on the inner surfaces of the two side walls **34** on two opposite sides of the second cover **35**B, and the two vertical walls **32** on two opposite sides of the second body **31**B are provided with the one or more sliders **57** corresponding to the two chute structures **50**, but are not limited thereto. Moreover, in an embodiment, each of the notches **49** may have a first opening M**1** at the outer surface of the peripheral wall **56** and a second opening M**2** at the inner surface of the peripheral wall **56**. A width of the first opening M**1** is greater than a width of the second opening M**2**, so that an inner surface of the notch **49** is an inclined surface for guiding the slider **57** to enter the chute **55** through the notch **49**. In addition, in an embodiment, the width of the second opening M**2** of the notch **49** may be slightly less than the maximum width of the slider **57**. After the slider **57** enters the chute **55**, a certain external force needs to be applied to deform the slider **57** slightly so that the slider **57** can pass through the second opening M**2** and leave the chute **55** through the first opening M**1**. Other details of the chute structures **50** (including the limiting elements **51**) and the sliders **57** in the second embodiment are as described in the first embodiment, and will not be described again here.

[0051] Please refer to FIG. **9**, which is a schematic diagram of the energy meter assembly **20** and the bracket **90** in FIG. **1** viewed from another angle. A second surface **42**B of the second body **31**B of the second energy meter **22** is provided with a holding structure **60**. The second surface **42**B and the first surface **41**B are two opposite surfaces of the second body **31**B. The holding structure **60** is configured to hold a first edge **91** of the bracket **90**. In an embodiment, the bracket **90** is a DIN rail, but is not limited thereto. Please refer to FIG. **10**, which is a cross-sectional view of the energy meter assembly **20** and the bracket **90** in FIG. **9** taken at a line **10-10'**. The holding structure **60** includes a first protrusion **61** and a second protrusion **62**. The first protrusion **61** is disposed on the second surface **42**B. The second protrusion **62** extends from the second surface **42**B and forms a space **65** with the second surface **42**B. The space **65** is configured to receive the first edge **91** of the bracket **90**. The first protrusion **61** is configured to abut against the first edge **91** of the bracket **90** such that the second protrusion **62** interferes with the first edge **91** of the bracket **90**, so as to clamp the first edge **91** of the bracket **90**.

[0052] As shown in FIG. **10**, in this embodiment, the first protrusion **61** is a semi-cylindrical bump, but it is not limited thereto, and may be a bump of other shapes. The second protrusion **62** has a first inclined surface **64**. The first inclined surface **64** faces the second surface **42**B of the second body **31**B, and is inclined away from the second surface **42**B in a direction toward the bracket **90** (i.e., a direction toward a holding element **70** described below). The first inclined surface **64** is configured to interfere with the first edge **91** of the bracket **90**. Specifically, when the first edge **91** of the bracket **90** is inserted into the space **65**, the first edge **91** of the bracket **90** is pushed up by the first protrusion **61** to abut against the first inclined surface **64**, so that the first edge **91** of the bracket **90** is firmly stuck in the holding structure **60**.

[0053] As shown in FIG. **10**, in an embodiment, a projection of the second protrusion **62** on the second surface **42**B does not overlap with a projection of the first protrusion **61** on the second surface **42**B. In other words, the first protrusion **61** is located outside the space **65**, so that the first inclined surface **64** optimally interferes with the first edge **91** of the bracket **90** and is more firmly stuck between the first protrusion **61** and the second protrusion **62**.

[0054] As shown in FIG. **9**, in an embodiment, the second housing **30**B further includes the holding element **70**. The holding element **70** is slidably disposed on the second surface **42**B of the second body **31**B and corresponds to the holding structure **60**. The holding element **70** is configured to hold a second edge **92** of the bracket **90**. The second edge **92** and the first edge **91** are two opposite sides of the bracket **90**. The holding structure **60** and the holding element **70** are configured to clamp the bracket **90**.

[0055] Please refer to FIG. **9** and FIG. **10**, after the first edge **91** of the bracket **90** is held by the holding structure **60**, the holding element **70** is slid toward the bracket **90** to hold the second edge **92** of the bracket **90**, so that the second energy meter **22** is installed on the bracket **90**. In an embodiment, a front end of the holding element **70** has a second inclined surface **74**. The second inclined surface **74** faces the second surface **42**B of the second body **31**B, and is inclined away from the second surface **42**B in a direction toward the bracket **90** (i.e., a direction toward the holding structure **60**). The second inclined surface **74** is configured to interfere with the second edge **92** of the bracket **90**, so that the holding element **70** can more firmly hold the second edge **92** of the bracket **90**.

[0056] Please refer to FIG. **11**, the second housing **30**B may include one or more guide rails **71** and one or more guide grooves **73**. Each of the guide rails **71** corresponds to one guide groove **73**. In this embodiment, two guide rails **71** are respectively provided on two opposite sides of the holding element **70**, and two corresponding guide grooves **73** are provided on the second surface **42**B of the second body **31**B. In other embodiments, two guide grooves **73** may be respectively provided on two opposite sides of the holding element **70**, and two corresponding guide rails **71** may be provided on the second surface **42**B of the second body **31**B. The guide rails **71** are slidably disposed in the guide grooves **73**, so that the holding element **70** is slidably disposed on the second surface **42**B of the second body **31**B.

[0057] Please refer to FIG. **11**, the second surface **42**B of the second body **31**B may be further provided with one or more first engaging elements **75**. Please refer to FIG. **12**, the holding element **70** has an elastic arm **77**, and the elastic arm **77** may be provided with a second engaging element **79**. The second engaging element **79** is configured to engage with any first engaging element **75** to fix a position of the holding element **70** on the second surface **42**B. In this embodiment, the first engaging element **75** is an engaging slot, and the second engaging element **79** is an engaging block mating the engaging slot. In other embodiments, the first engaging element **75** may be an engaging block, and the second engaging element **79** may be an engaging slot mating the engaging block. In this embodiment, the engaging block has a semi-cylindrical structure, but is not limited to this and may have other shapes.

[0058] Please refer to FIG. **10** and FIG. **11**, in this embodiment, the second surface **42**B may be provided with two first engaging elements **75**. The second engaging element **79** can engage with either one of the first engaging elements **75**, so that the holding element **70** can be held at different positions on the second surface **42**B. Please refer to FIG. **10**, when the second engaging element **79** is engaged with the first engaging element **75** close to the bracket **90**, the holding element **70** is held at a first position and holds the second edge **92** of the bracket **90**. When the second engaging element **79** is engaged with the first engaging element **75** away from the bracket **90**, the holding element **70** is held at a second position and is separated from the second edge **92** of the bracket **90**, so as to facilitate separating the bracket **90** from the second energy meter **22**.

[0059] Please refer to FIG. **2** and FIG. **4**, in the first energy meter **21**, a first surface **41**A of the first body **31**A is also provided with one or more vertical walls **32**, and the first cover **35**A also has one or more side walls **34**. The first housing **30**A of the first energy meter **21** also includes one or more sliders **57** and one or more chute structures **50**. The chute structure **50** is disposed on one of the vertical wall **32** and the side wall **34**, and the slider **57** is disposed on the other one of the vertical wall **32** and the side wall **34**. The first body **31**A further has a fifth surface **45**A and a sixth surface **46**A opposite to the fifth surface **45**A. Please refer to FIG. **1**, FIG. **2**, and FIG. **5** to FIG. **8**, the slider **57** can slide to different positions, so that the first cover **35**A exposes the terminal block **23** at the fifth surface **45**A, exposes the third connector **24** at the sixth surface **46**A, or covers the terminal block **23** and the third connector **24**. Other details of the sliders **57** and the chute structures **50** of the first energy meter **21** are as described above for the second energy meter **22** and will not be described again here.

[0060] Please refer to FIG. **9** to FIG. **12**, in the first energy meter **21**, a second surface **42**A of the first body **31**A of is also provided with the holding structure **60**, the holding element **70**, the one or more guide rails **71** and the one or more guide grooves **73**, the details of which are as described above for the second energy meter **22** and will not be described again here.

[0061] In sum, a housing of an energy meter of the present disclosure includes a body and a cover slidably disposed on the body. The cover can slide upward or downward relative to the body to expose a terminal block and/or a connector disposed on the body, thereby facilitating a user to connect external devices to the energy meter through wires. Furthermore, the cover is slidably disposed on the body through one or more sliders and one or more chute structures. Each of the chute structures includes a chute and a peripheral wall surrounding the chute. The peripheral wall is provided with one or more limiting elements configured to limit positions of the sliders in the chute, so that the sliders can stay at specific positions. Moreover, a back surface of the body may be provided with a holding structure and a slidable holding element to facilitate the user to mount the energy meter on a bracket. In addition, a side surface of the body may be provided with a boss or a recess. Through a combination of the boss and the recess, multiple energy meters can be combined into an energy meter assembly.

[0062] Although the present disclosure has been described with the above embodiments, the above embodiments are not intended to limit the present disclosure. Any person skilled in the art can make various changes and modifications without departing from the core and scope of the present disclosure. Therefore, the protective scope of the present disclosure shall be determined by the attached claims.

What is claimed is:

1. A housing for an energy meter, comprising:

a body having a first surface and a vertical wall substantially perpendicular to the first surface;

a cover having a side wall; and

a chute structure and a slider, wherein the chute structure comprises a chute and a peripheral wall surrounding the chute, and the peripheral wall is provided with a first limiting element;

wherein the chute structure is disposed on one of the vertical wall and the side wall, the slider is disposed on the other one of the vertical wall and the side wall, the slider is slidably disposed in the chute, the first limiting element is configured to limit a position of the slider, and the cover is slidably disposed on the body through the slider and the chute structure.

2. The housing for the energy meter according to claim **1**, wherein the first limiting element is close to an end of the chute, when the slider is limited between the first limiting element and the end of the chute, a front edge of the cover is in front of a front edge of the body, or a rear edge of the cover is behind a rear edge of the body.

3. The housing for the energy meter according to claim **1**, wherein the peripheral wall of the chute structure is further provided with a second limiting element and a third limiting element, when the slider is limited between the second limiting element and the third limiting element, a front edge of the cover is substantially aligned with a front edge of the body, or a rear edge of the cover is substantially aligned with a rear edge of the body.

4. The housing for the energy meter according to claim **1**, wherein the peripheral wall is further provided with a notch, and the notch communicates with the chute and is configured to allow the slider to pass.

5. The housing for the energy meter according to claim **4**, wherein the notch has a first opening at an outer surface of the peripheral wall and a second opening at an inner surface of the peripheral wall, and a width of the first opening is greater than a width of the second opening.

6. The housing for the energy meter according to claim **1**, wherein the body further comprises a second surface, the second surface is provided with a holding structure, the holding structure comprises a first protrusion and a second protrusion, the first protrusion is disposed on the second surface, the second protrusion extends from the second surface and forms a space with the second surface, the space is configured to receive a first edge of a bracket, and the first protrusion is configured to abut against the first edge such that the second protrusion interferes with the first edge.

7. The housing for the energy meter according to claim **6**, wherein the second protrusion has a first inclined surface, and the first inclined surface faces the second surface and is configured to interfere with the first edge of the bracket.

8. The housing for the energy meter according to claim **6**, wherein a projection of the second protrusion on the second surface does not overlap with a projection of the first protrusion on the second surface.

9. The housing for the energy meter according to claim **6**, further comprising a holding element, a guide rail, and a guide groove, wherein the guide rail is disposed on one of the holding element and the second surface, the guide groove is disposed on the other one of the holding element and the second surface, the guide rail is slidably disposed in the guide groove, so that the holding element is slidably disposed on the second surface, and the holding element is configured to hold a second edge of the bracket.

10. The housing for the energy meter according to claim **9**, wherein a front end of the holding element has a second inclined surface, and the second inclined surface faces the second surface and is configured to interfere with the second edge of the bracket.

11. The housing for the energy meter according to claim **9**, wherein the second surface is further provided with a first engaging element, the holding element has an elastic arm, the elastic arm is provided with a second engaging element, and the second engaging element is configured to engage with the first engaging element to fix a position of the holding element on the second surface.

12. The housing for the energy meter according to claim **1**, wherein the body further comprises a third surface, the third surface is provided with a boss, and an outer wall of the boss is provided with a first latching element.

13. The housing for the energy meter according to claim **12**, wherein the body further comprises a fourth surface, the fourth surface is provided with a recess that matches the boss, and an inner wall of the recess is provided with a second latching element that matches the first latching element.

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